

among many networks and gadgets. When rows and partitions are deleted or added to the network, they are adjusted across it.

Cassandra is a unique blockchain database because of its numerous significant characteristics. Each node in the cluster has a row and a partition. Because every cluster is both a client and a server, there is no master cluster. Cassandra is a fault-tolerant system because it is distributed and does not have a single point of control.

If one of the nodes in the cluster is attacked, it doesn't bring down the entire system since each of the nodes has a copy of the database. Because the data from the node that is down is still stored on the cluster's other nodes, there is no risk to the information on it. The rows, tables, and columns distinguish Cassandra from other relational databases. Cassandra does not query data using the SQL language as it uses its own query language Cassandra Query Language (CQL).

ChainifyDB

(<https://www.chainifydb.com/>)

ChainifyDB is a blockchain based database management system. Databases may be integrated into the ChainifyDB network and their records synchronised across the network. Once a record is added to a database, ChainifyDB notifies all other database nodes of the new information. They reach an agreement and the record is written to the databases, making it decentralised, unchangeable, and open.

Unlike other blockchain based databases, ChainifyDB has its own database/storage space for each block it is part of. A blockchain layer is plugged into the databases offered by ChainifyDB, with security, privacy and integrity.

The ChainifyDB network uses strong encryption to transmit data between its connected databases. It is possible to execute ChainifyDB's basic components and administrative configuration via a Web frontend. As it doesn't require a large number of tools to set up, it is more user friendly than other databases. ChainifyDB

may be inserted into any database without disrupting the apps that are running on it.

CovenantSQL

(<https://covenantsql.io/>)

CovenantSQL is a database used widely for blockchain based decentralised applications. As stated on the official website, CovenantSQL intends to make DApp development easier by connecting idle storage resources via a consensus mechanism. Like Ethereum, CovenantSQL provides a foundation over which decentralised applications may be built. This database has a wide range of applications, including asset management and IoT integration.

CovenantSQL is distributed through a P2P network, just like the blockchain technology it employs. Querying databases with SQL is the most common database operation. The immutability of the database is ensured by CovenantSQL's blockchain. Before they can be committed to the database, all records must be verified by every node in the network.

Modex Blockchain Database (BCDB)

(<https://modex.tech/blockchain-database-product/>)

Modex Blockchain Database (BCDB) offers a simple plug-and-play approach to blockchain development for enterprises. Client applications and databases are separated by Modex BCDB. It can be used with various blockchain frameworks since it is adaptable and easy to integrate. The Hyperledger Sawtooth framework and the Tendermint protocol are now being used in the network and consensus protocols of this platform.

Modex BCDB can support multiple databases. It can smoothly operate and sync data between a node that uses MongoDB and one that uses MySQL, without any configuration or porting to a supported database. Modex BCDB is capable of managing data flawlessly and without sacrificing security in anyway. Nodes in the Modex BCDB network can

either be fully or partially accessible to all data, depending on their API requests.

Postchain

(<https://postchain-docs.readthedocs.io/en/latest/>)

Like Ethereum and Hyperledger, Postchain has a blockchain architecture with a distributed network of nodes. It uses a proof-of-authority consensus process to retain a collection of data. Postchain keeps this data in an SQL database, and every node on the Postchain network has a validator. Transactions are encrypted and signed before being sent. A group of validators works together to verify the message's provenance and source. This synchronisation is carried out to ensure that all network nodes have the same database state.

ProvenDB

(<https://www.provendb.com/>)

ProvenDB is a MongoDB database that incorporates blockchain technology. Records in ProvenDB can never be changed or removed once they are entered. All types of sensitive data can be stored in a highly secure blockchain database, including financial records that cannot be changed as well as intellectual property, legal papers and public records. ProvenDB delivers a secure database that is encrypted and resistant to hacking.

Blockchain based dedicated database engines can help to develop real-time and secure applications for smart cities, e-governance, supply chain management, education based ERP, etc. The performance of such blockchain based database systems with respect to energy, security, privacy, etc, can be analysed effectively by using different programming platforms.  END

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